Lab Report - Lab Unit 1

Use Case Description

Software and Process Notations

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Use Case Name

Automated Valet Parking & Charging (Compliance and Data Privacy Management)

Description

Customers arrive at the entry terminal, the license plate is recognized and a session is created. The driver now has the choice to bind their identity to the session via a bank card through a payment token. If the driver cannot or does not want to use a card, a printed ticket with a QR code and an alphanumeric code is issued instead. The driver can optionally request charging, and the car is then parked automatically.

Preconditions

- The facility is operational with at least one free space (and a free charger if charging is requested).
- The entry and exit terminals are functional, with a working card reader, a connection to the payment provider and ticket paper available.
- The vehicle is a passenger car within the size and weight limits.
- The privacy notice and support hours are visible at the terminal.

Postconditions

- The car was parked, charged if requested, and handed over to the identified driver.
- Payment is settled.
- The plate, card token or ticket token hash, and all other session identifiers are deleted. Only an anonymous billing record (session ID, timestamps, amount, kWh) remains for the legally required period.
- Anonymized data is available for statistics.

Frequency of Use

Continuous throughout the day, with peaks at commuting hours and events.

Normal Course of Events

1. The driver drives into the entry bay and stops.
2. The system checks that the vehicle is eligible (type and size).
3. The system captures the plate image, extracts the plate text and immediately discards the image.
4. The driver confirms the recognized plate.
5. The system displays the tariff, support hours and a short privacy notice.
6. The driver optionally selects charging.
7. The driver taps or inserts a bank card.
8. The system requests a payment token (and a pre-authorization reference if applicable) from the payment provider and stores only the token.
9. The system creates the session, linking plate and card token.
10. The driver leaves the bay, and the system verifies that it is empty.
11. The system assigns a space, parks the car and starts charging if requested. The session status becomes "parked".
12. The driver taps the same card at the exit terminal.
13. The system matches the card token to the active session.
14. The system brings the car to the handover bay and stops charging.
15. The system recognizes the plate again and matches it with the session as a second check.
16. The system calculates the fee and settles payment with the card.
17. The driver takes the car.
18. The system closes the session and deletes plate and card token, keeping only the anonymous billing record.

Alternate Course

Ticket instead of card (fallback) In step 7 the driver chooses "no card" (for example cash customer, foreign cards, or a preference for anonymity). The system generates a random token, stores only its hash, creates the session, and prints a ticket (QR code, alphanumeric code, ticket number, entry time, support hours). The course then continues at step 12.

Retrieval with ticket In step 12 the driver scans the QR code. If it is unreadable, the driver types the alphanumeric code together with the plate. The course continues at step 14.

Driver leaves immediately after check-in The system cancels the session, releases any pre-authorization, and deletes all data. Nothing is charged (or a short free grace period applies).

Driver rejects the recognized plate (step 4) The driver corrects it manually or aborts. If aborted, all data is deleted.

Charging not requested The charging steps are skipped.

Exceptions

Plate not recognized The system retries up to 3 times, then offers manual plate entry with confirmation. The system compares the typed plate with the captured image and calculates a confidence value. If the confidence is at or above the threshold, the plate is accepted and the course continues at step 4 of the normal course. If the confidence is below the threshold, the system refuses entry, discards the image, and instructs the driver to leave the entry bay. The driver can clean the plate or fix the obstruction outside and try again. No session is created and nothing is stored.

Vehicle not eligible (for example a motorcycle or oversized vehicle) Entry is refused before any session is created, and nothing is stored.

No free space or charger The driver is informed before a session is created. Parking without charging may be offered.

Card declined or not readable at entry The system allows one more attempt, then offers another card or the ticket path.

Payment provider unavailable The system offers the ticket path so entry remains possible. Payment is settled at exit. If the ticket path is also unavailable, entry is refused and the driver leaves the bay.

Ticket printer out of paper or defective (ticket path only) No session is created, and the driver is offered the card path. The system sends a maintenance alert to the operator.

Obstruction or person detected in the bay Parking is paused until the bay is clear.

Plate mismatch at handover (step 15) The car is not released, and the driver is directed to support.

Too many failed attempts at the exit terminal The session is temporarily locked and the driver is directed to support.

Card and ticket lost, replaced or unusable The terminal tells the driver to contact support and shows the support hours. At the desk, staff visually check the original ID and the original registration document, and confirm that the plate matches the parked car. If the registered holder differs from the driver, an authorization letter is required or the release is declined. Staff then release the session manually, and the plate match at the handover bay still applies. Nothing is scanned or copied, and the log records only session reference, time, staff member, documents checked and outcome.

System or terminal failure No session is created, and existing sessions stay intact. At entry, the system refuses entry and instructs the driver to leave the bay. At the exit terminal, the driver is directed to support.

Includes

- Recognize License Plate (entry and handover)
- Bind Payment Card to Session
- Park Vehicle Automatically in Facility
- Charge Vehicle
- Settle Payment
- Retrieve Vehicle

Special Requirements

GDPR

- The plate image is held only for the duration of plate extraction and, if needed, the manual-entry confidence check. It is discarded immediately afterwards, whether the plate is accepted or rejected.
- The plate is processed only for parking, charging and billing (lawful basis: performance of the contract). The identity check by customer support when card and ticket are lost rests on legitimate interest, backed by a documented balancing test.

- Signage and a privacy notice appear before capture.
- Images are discarded right after plate extraction.
- Only the payment token is stored, never the card number. The payment provider is a processor and needs a data processing agreement.
- Ticket tokens are random, never derived from the plate, and the QR code contains no personal data.
- Plate and token are deleted at session closure, and billing records contain no plate data.
- Data is stored on the company server with access control and encryption, and backups follow the deletion rules.
- No biometric processing, and no copies of identity documents.
- Statistics use only anonymized or aggregated data.
- A DPIA (Data Protection Impact Assessment (Art. 35 GDPR)) is carried out before operation. That is a risk analysis regarding systematic monitoring of people recorded on the facility, automated recognition technology and large-scale processing.

Other

- No registration is required, and both card and ticket paths are always offered.
- Card handling follows PCI DSS (Payment Card Industry Data Security Standard), by delegating card data entirely to the payment provider.
- Retrieval with a valid card or ticket is available 24/7. Lost credential recovery is available only during support hours, which are shown on the terminal and tickets.
- The alphanumeric code avoids easily confused characters (0/O, 1/I/l).
- Session creation completes within a defined time (for example under 10 seconds).

Assumptions

- The entry area allows a driver to turn around or leave without a session, so a refused driver is never stuck in the bay.
- The driver has a passenger car with a valid, legible license plate (motorcycles and other vehicle types are out of scope).
- The car fits the size and weight limits of the facility.
- The driver has a bank card or accepts the ticket path, and keeps the card or ticket for the whole stay.

- For recovery, the driver can come to the desk in person with ID and registration document. If the papers are locked in the parked car, staff may fetch them under supervision after the ID check.
- The driver accepts the tariff and terms at the terminal.

Statement on the Use of AI

In preparing this lab report, Claude Code (Anthropic, Claude Opus 5.5, token based billing) was used as a generative AI tool exclusively to assist with the following tasks:

- creating and designing the LaTeX layout,
- checking spelling and grammar,
- research.

The LaTeX template of this report, including this statement, was created with the assistance of the AI tool and reviewed and adapted by the lab members. Apart from that, all content, results and conclusions were developed independently; no AI-generated text was adopted. The tool was used in line with the principles of *responsible AI use* (transparency, verifiability, human oversight). Sources found through research were checked for accuracy independently. All content was reviewed, validated and finally approved by the lab members (cf. FH Salzburg AI guideline *KI-Richtlinie*, version 1.0, ch. 1.3 and 2.1.2).